4. Banking Loan Approval Dataset
Question 4
A bank wants to analyze loan approval decisions based on income, credit score, loan amount, employment type, and approval status. Perform EDA to identify patterns related to loan approval.

Customer_ID	Income	Credit_Score	Loan_Amount	Employment_Type	Loan_Status
C01	45000	720	200000	Salaried	Approved
C02	25000	580	150000	Self-employed	Rejected
C03	60000	750	300000	Salaried	Approved
C04	22000	560	120000	Unemployed	Rejected
C05	50000	710	250000	Salaried	Approved
C06	30000	600	180000	Self-employed	Rejected
C07	70000	780	350000	Salaried	Approved
C08	28000	590	160000	Self-employed	Rejected
C09	55000	730	270000	Salaried	Approved
C10	24000	570	130000	Unemployed	Rejected
C11	65000	760	320000	Salaried	Approved
C12	35000	620	200000	Self-employed	Rejected
C13	48000	700	240000	Salaried	Approved
C14	26000	585	140000	Unemployed	Rejected
C15	75000	790	400000	Salaried	Approved
C16	32000	610	190000	Self-employed	Rejected
C17	52000	725	260000	Salaried	Approved
C18	27000	575	150000	Self-employed	Rejected
C19	68000	770	330000	Salaried	Approved
C20	23000	555	125000	Unemployed	Rejected

Answer:
```
# ======================================
# Banking Loan Approval EDA (4 Charts)
# ======================================

import pandas as pd
import matplotlib.pyplot as plt

# Load Dataset
df = pd.read_csv("banking_loan_dataset.csv")

# -----------------------------
# Basic Information
# -----------------------------
print("Dataset Shape:", df.shape)

print("\nMissing Values")
print(df.isnull().sum())

print("\nStatistical Summary")
print(df.describe())
```

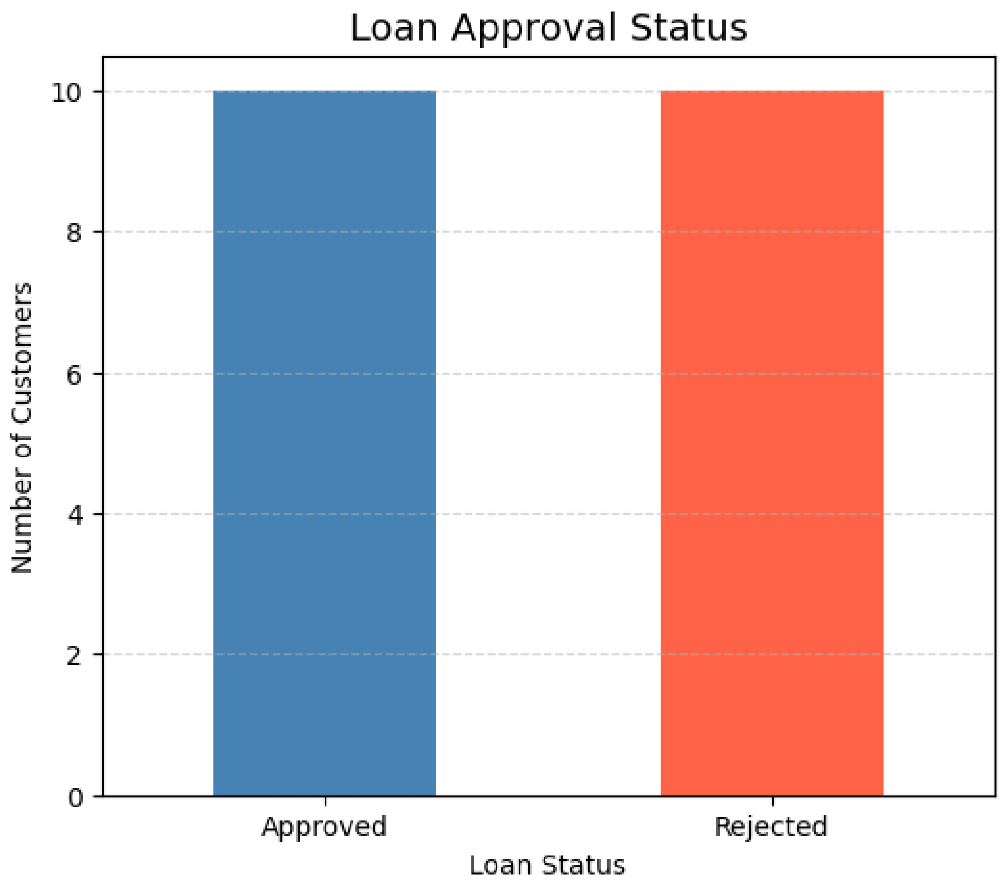
```
# ==========================================
# Chart 1 : Loan Approval Status
# ==========================================
plt.figure(figsize=(6,5))

df["Loan_Status"].value_counts().plot(
    kind="bar",
    color=["steelblue","tomato"]
)

plt.title("Loan Approval Status", fontsize=14)
plt.xlabel("Loan Status")
plt.ylabel("Number of Customers")

plt.xticks(rotation=0)
plt.grid(axis='y', linestyle='--', alpha=0.5)
plt.show()

# ==========================================
# Chart 2 : Average Income by Loan Status
# ==========================================
plt.figure(figsize=(6,5))

df.groupby("Loan_Status")["Income"].mean().plot(
    kind="bar",
    color=["green","orange"]
)

plt.title("Average Income by Loan Status", fontsize=14)
plt.xlabel("Loan Status")
plt.ylabel("Average Income")

plt.xticks(rotation=0)
plt.grid(axis='y', linestyle='--', alpha=0.5)
plt.show()

# ==========================================
# Chart 3 : Credit Score vs Loan Amount
# ==========================================
plt.figure(figsize=(7,5))

colors = df["Loan_Status"].map({
    "Approved":"green",
    "Rejected":"red"
})

plt.scatter(
    df["Credit_Score"],
    df["Loan_Amount"],
    c=colors,
```

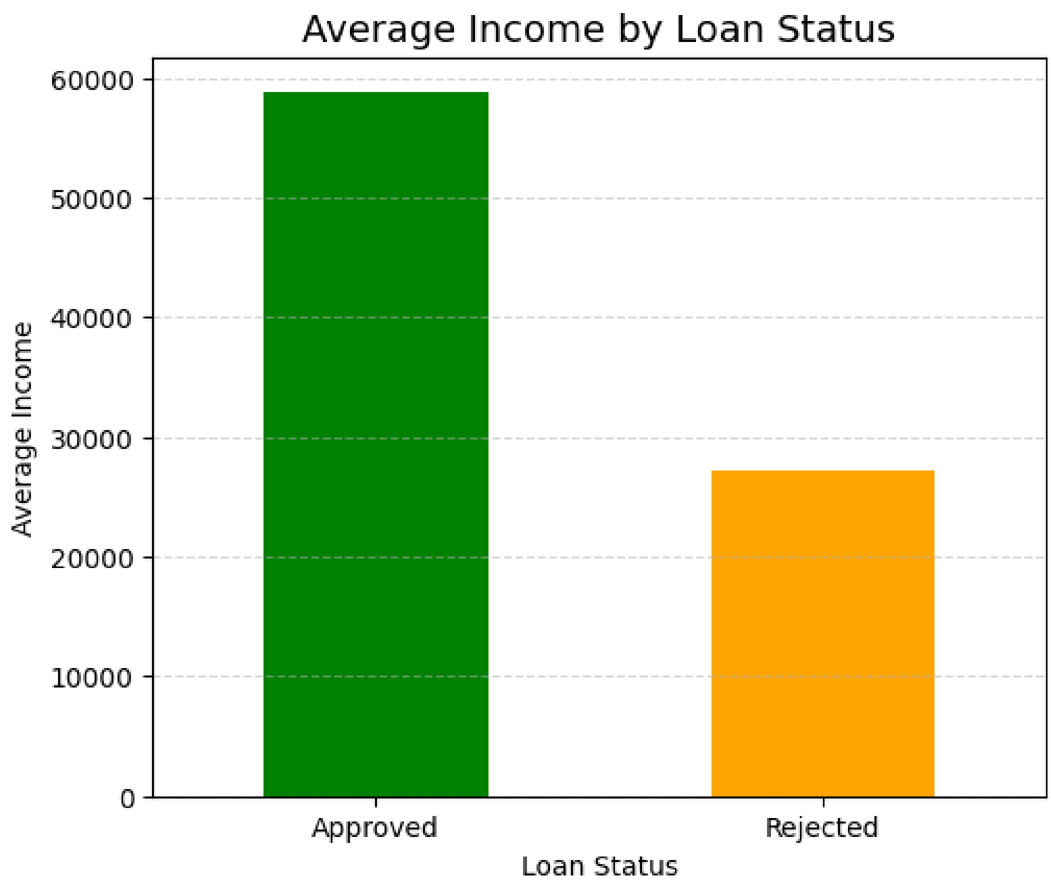
```
    s=100
)

plt.title("Credit Score vs Loan Amount", fontsize=14)
plt.xlabel("Credit Score")
plt.ylabel("Loan Amount")

plt.grid(True)

from matplotlib.lines import Line2D

legend = [
    Line2D([0],[0],marker='o',color='w',
           label='Approved',
           markerfacecolor='green',
           markersize=10),
    Line2D([0],[0],marker='o',color='w',
           label='Rejected',
           markerfacecolor='red',
           markersize=10)
]

plt.legend(handles=legend)
plt.show()

# ========================================
# Chart 4 : Correlation Heatmap
# ========================================
corr = df[["Income","Credit_Score","Loan_Amount"]].corr()

plt.figure(figsize=(6,5))

plt.imshow(corr, cmap="coolwarm")
plt.colorbar()

plt.xticks(range(len(corr.columns)), corr.columns, rotation=45)
plt.yticks(range(len(corr.columns)), corr.columns)

plt.title("Correlation Between Loan Variables", fontsize=14)

for i in range(len(corr.columns)):
    for j in range(len(corr.columns)):
        plt.text(j, i,
                 f"{corr.iloc[i,j]:.2f}",
                 ha='center',
                 va='center')

plt.tight_layout()
plt.show()
```

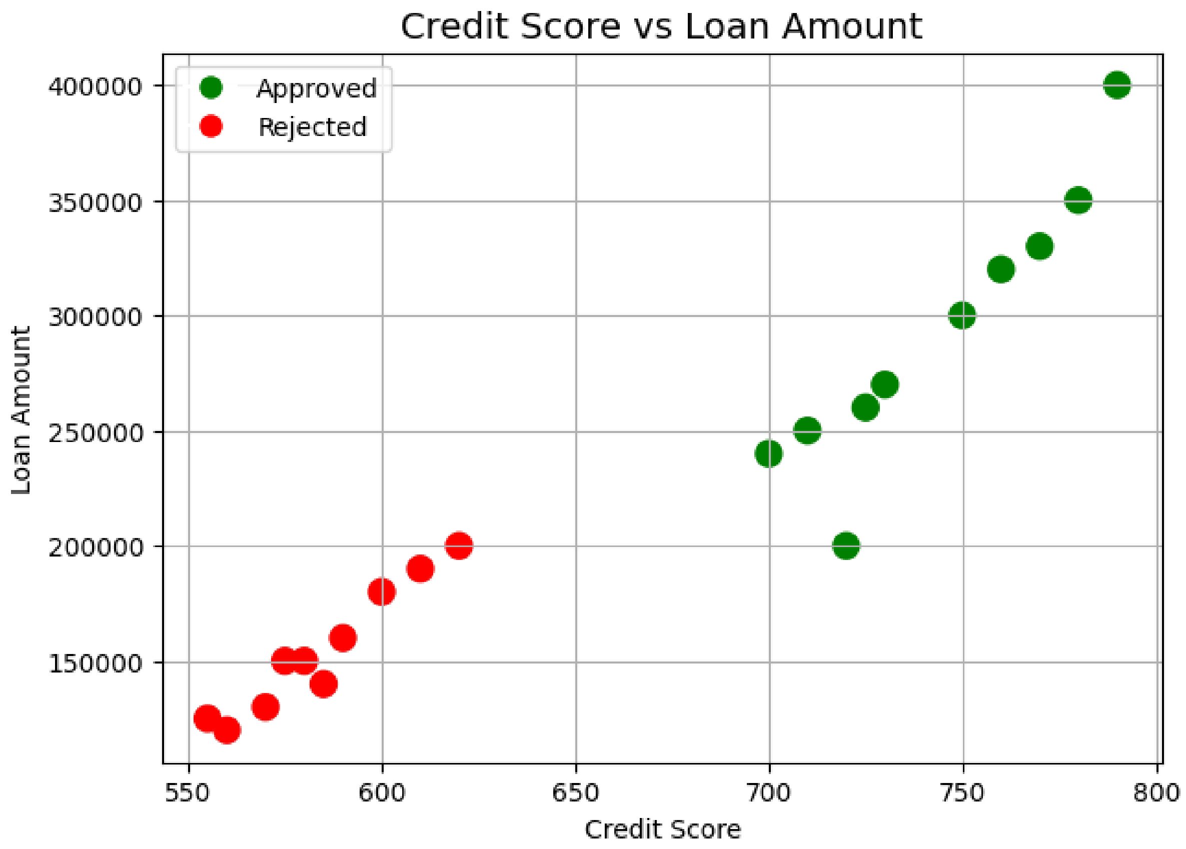
```
# ==========================================
# Approved & Rejected Customers
# ==========================================

print("\nApproved Customers")
print(df[df["Loan_Status"]=="Approved"][["Customer_ID","Income","Credit_Score"]])

print("\nRejected Customers")
print(df[df["Loan_Status"]=="Rejected"][["Customer_ID","Income","Credit_Score"]])

# ==========================================
# Final Result
# ==========================================

print("\n========== FINAL RESULT ==========")
print("1. Customers with higher income and higher credit scores are more likely to receive loan approval.")
print("2. Salaried customers have the highest number of approved loans.")
print("3. Approved customers generally receive larger loan amounts than rejected customers.")
print("4. Income, Credit Score, and Loan Amount show a positive correlation.")
print("5. No missing values or duplicate records are present, making the dataset suitable for further loan prediction analysis.")
```

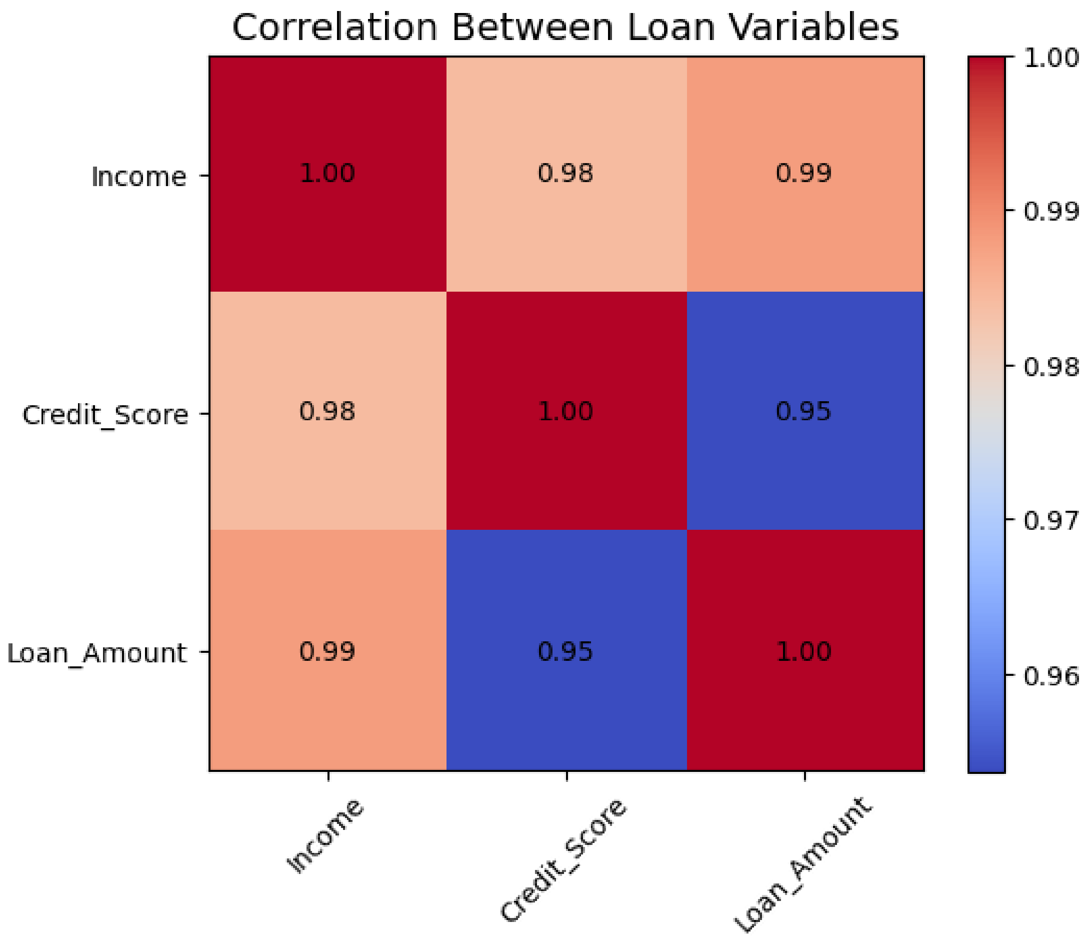
output

```
Dataset Shape: (20, 6)

Missing Values
Customer_ID       0
Income            0
Credit_Score      0
Loan_Amount       0
Employment_Type   0
Loan_Status       0
dtype: int64

Statistical Summary
             Income  Credit_Score    Loan_Amount
count     20.000000     20.000000      20.000000
mean   43000.000000    664.000000  223250.000000
std    17923.815383     85.526235   83733.207272
min    22000.000000    555.000000  120000.000000
25%    26750.000000    583.750000  150000.000000
50%    40000.000000    660.000000  200000.000000
75%    56250.000000    735.000000  277500.000000
max    75000.000000    790.000000  400000.000000
```


Loan Approval Status
Number of Customers
10
8
6
4
2
0
Approved
Rejected
Loan Status


Average Income by Loan Status
Average Income
60000
50000
40000
30000
20000
10000
0
Approved
Rejected
Loan Status


Credit Score vs Loan Amount
Approved
Rejected
Loan Amount
Credit Score
400000
350000
300000
250000
200000
150000
550
600
650
700
750
800


Correlation Between Loan Variables
Income
Credit_Score
Loan_Amount
1.00
0.98
0.99
0.98
1.00
0.95
0.99
0.95
1.00
1.00
0.99
0.98
0.97
0.96

Approved Customers

	Customer_ID	Income	Credit_Score
0	C01	45000	720
2	C03	60000	750
4	C05	50000	710
6	C07	70000	780
8	C09	55000	730
10	C11	65000	760
12	C13	48000	700
14	C15	75000	790
16	C17	52000	725
18	C19	68000	770

Rejected Customers

	Customer_ID	Income	Credit_Score
1	C02	25000	580
3	C04	22000	560
5	C06	30000	600
7	C08	28000	590
9	C10	24000	570
11	C12	35000	620
13	C14	26000	585
15	C16	32000	610
17	C18	27000	575
19	C20	23000	555

========== FINAL RESULT ==========

1. Customers with higher income and higher credit scores are more likely to receive loan approval.
2. Salaried customers have the highest number of approved loans.
3. Approved customers generally receive larger loan amounts than rejected customers.
4. Income, Credit Score, and Loan Amount show a positive correlation.
5. No missing values or duplicate records are present, making the dataset suitable for further loan prediction analysis.